

Handling Missing Values and Outliers:

➤ Impute mean for the following columns, and round off to the nearest integer if required: WarehouseToHome, HourSpendOnApp, OrderAmountHikeFromLastYear, DaySinceLastOrder.

```
5660 • SELECT
5661     ROUND(AVG(WarehouseToHome)) AS avg_WarehouseToHome,
5662     ROUND(AVG(HourSpendOnApp)) AS avg_HourSpendOnApp,
5663     ROUND(AVG(OrderAmountHikeFromLastYear)) AS avg_OrderAmountHikeFromLastYear,
5664     ROUND(AVG(DaySinceLastOrder)) AS avg_DaySinceLastOrder
5665 FROM customer_churn;
```

Result Grid

Filter Rows:

Export:

Wrap Cell Content:

	avg_WarehouseToHome	avg_HourSpendOnApp	avg_OrderAmountHikeFromLastYear	avg_DaySinceLastOrder
▶	16	3	16	5

➤ Impute mode for the following columns: Tenure, CouponUsed, OrderCount.

```
5721
5722 select tenure,
5723 CouponUsed,
5724 OrderCount from
5725 customer_churn where tenure = null or couponUsed = null or ordercount = null
```

Result Grid	Filter Rows:	Export:	Wrap Cell Content:
	tenure	CouponUsed	OrderCount

➤ Handle outliers in the 'WarehouseToHome' column by deleting rows where the values are greater than 100.

```
14 select count(*) from customer_churn where warehouseToHome > 100
15
16 delete from customer_churn where warehouseToHome > 100
```

Result Grid

Filter Rows:

Export:

Wrap Cell Content:

	count(*)
▶	0

Dealing with Inconsistencies:

➤ Replace occurrences of "Phone" in the 'PreferredLoginDevice' column and "Mobile" in the 'PreferredOrderCat' column with "Mobile Phone" to ensure uniformity.

```

11
12 update customer_churn set orderCount = '2' where orderCount is null
13
14 select count(*) from customer_churn where PreferredLoginDevice = 'Mobile Phone' or PreferredOrderCat = 'Mobile Phone'
15
16 update customer_churn set PreferredLoginDevice = 'Mobile Phone'
17 where PreferredLoginDevice = 'Phone'
18
19 update customer_churn set PreferredOrderCat = 'Mobile Phone'
20 where PreferredOrderCat = 'Mobile'
21

```

Result Grid

count(*)
4590

➤ Standardize payment mode values: Replace "COD" with "Cash on Delivery" and "CC" with "Credit Card" in the PreferredPaymentMode column.

```

24 select count(*) from customer_churn where PreferredPaymentMode in ('COD','CC')
25
26 update customer_churn set PreferredPaymentMode = 'Cash on Delivery' WHERE PreferredPaymentMode = 'COD'
27
28 update customer_churn set PreferredPaymentMode = 'Credit Card' WHERE PreferredPaymentMode = 'CC'
29

```

Result Grid

count(*)
0

Data Transformation:

Column Renaming:

➤ Rename the column "PreferedOrderCat" to "PreferredOrderCat".

➤ Rename the column "HourSpendOnApp" to "HoursSpentOnApp".

SQL File 4* SQL File 5* SQL File 6* SQL File 5 SQL File 5 E-Commerce Customer churn db* SQL File

```

1 SELECT PreferredOrderCat, HoursSpentOnApp FROM CUSTOMER_CHURN
2
3 ALTER TABLE CUSTOMER_CHURN
4 RENAME COLUMN PreferedOrderCat TO PreferredOrderCat
5
6 ALTER TABLE CUSTOMER_CHURN
7 RENAME COLUMN HourSpendOnApp TO HoursSpentOnApp
8

```

Result Grid

PreferredOrderCat	HoursSpentOnApp
Laptop & Accessory	3
Mobile Phone	3
Mobile Phone	2
Laptop & Accessory	2
Mobile Phone	3

Creating New Columns:

➤ Create a new column named 'ComplaintReceived' with values "Yes" if the corresponding value in the 'Complain' is 1, and "No" otherwise.

SQL Query Editor showing SQL commands for the `CUSTOMER_CHURN` table:

```

1 ALTER TABLE CUSTOMER_CHURN
2 ADD COLUMN ComplaintReceived varchar(30)
3
4 update customer_churn set ComplaintReceived = 'Yes' where complain = '1'
5
6 update customer_churn set ComplaintReceived = 'No' where complain = '0'
7
8 select complain, ComplaintReceived, customer_churn.* from customer_churn
9
10

```

Result Grid:

complain	ComplaintReceived	CustomerID	Churn	Tenure	PreferredLoginDevice	CityTier	WarehouseToHome	PreferredPaymentMode	Gender	HoursSpentOnApp	NumberOfDeviceRegistered
1	Yes	50001	1	4	Mobile Phone	3	6	Debit Card	Female	3	3
1	Yes	50002	1	1	Mobile Phone	1	8	UPI	Male	3	4
1	Yes	50003	1	1	Mobile Phone	1	30	Debit Card	Male	2	4
0	No	50004	1	0	Mobile Phone	3	15	Debit Card	Male	2	4

➤ Create a new column named 'ChurnStatus'. Set its value to "Churned" if the corresponding value in the 'Churn' column is 1, else assign "Active".

Column Dropping:

SQL Query Editor showing SQL commands for the `CUSTOMER_CHURN` table:

```

1 ALTER TABLE CUSTOMER_CHURN
2 ADD COLUMN ChurnStatus varchar(30)
3
4 update customer_churn set ChurnStatus = 'Churned' where Churn = '1'
5
6 update customer_churn set ChurnStatus = 'Active' where Churn = '0'
7
8 select ChurnStatus, Churn, customer_churn.* from customer_churn
9
10

```

Result Grid:

ChurnStatus	Churn	CustomerID	Churn	Tenure	PreferredLoginDevice	CityTier	WarehouseToHome	PreferredPaymentMode	Gender	HoursSpentOnApp	NumberOfDeviceRegistered	PreferredOrderCat	SatisfactionScore
Churned	1	50025	1	14	Computer	1	14	Debit Card	Male	2	5	Fashion	2
Churned	1	50026	1	0	Mobile Phone	1	15	Credit Card	Female	2	3	Mobile Phone	3
Active	0	50027	0	8	Mobile Phone	3	6	E wallet	Male	3	3	Fashion	3
Active	0	50028	0	1	Mobile Phone	3	12	E wallet	Male	2	3	Laptop & Accessory	5

➤ Drop the columns "Churn" and "Complain" from the table.

SQL Query Editor showing SQL commands for the `CUSTOMER_CHURN` table:

```

1 ALTER TABLE CUSTOMER_CHURN
2 DROP COLUMN Churn
3
4 ALTER TABLE CUSTOMER_CHURN
5 DROP COLUMN Complain
6
7 select * from customer_churn
8
9

```

Result Grid:

CustomerID	Tenure	PreferredLoginDevice	CityTier	WarehouseToHome	PreferredPaymentMode	Gender	HoursSpentOnApp	NumberOfDeviceRegistered	PreferredOrderCat	SatisfactionScore
50001	4	Mobile Phone	3	6	Debit Card	Female	3	3	Laptop & Accessory	2
50002	1	Mobile Phone	1	8	UPI	Male	3	4	Mobile Phone	3
50003	1	Mobile Phone	1	30	Debit Card	Male	2	4	Mobile Phone	3
50004	0	Mobile Phone	3	15	Debit Card	Male	2	4	Laptop & Accessory	5

Data Exploration and Analysis:

➤ Retrieve the count of churned and active customers from the dataset.

```

1 • select churnstatus,count(*) from customer_churn
2   group by churnstatus

```

Result Grid			Filter Rows:	Export:	Wrap Cell Content:
	churnstatus	count(*)			
▶	Churned	948			
	Active	4680			

➤ Display the average tenure and total cashback amount of customers who churned.

```

1 • select * from customer_churn
2
3 • select avg(tenure) as Average_tenure, sum(cashbackAmount) as totalcashbackamount from customer_churn where churnstatus = 'Churned'

```

Result Grid			Filter Rows:	Export:	Wrap Cell Content:
	Average_tenure	totalcashbackamount			
▶	3.1762	152030			

➤ Determine the percentage of churned customers who complained.

```

1 • SELECT
2   (SELECT COUNT(*) FROM customer_churn WHERE churnstatus = 'Churned' AND ComplaintReceived = 'Yes')
3   * 100.0 /
4   (SELECT COUNT(*) FROM customer_churn WHERE churnstatus = 'Churned')
5   AS Complaint_Percentage;

```

Result Grid			Filter Rows:	Export:	Wrap Cell Content:
	Complaint_Percentage				
▶	53.58650				

➤ Identify the city tier with the highest number of churned customers whose preferred order category is Laptop & Accessory.

Limit to 1000 rows

```

1 • select * from customer_churn
2
3 ✖ select cityTier, count(*)
4   from customer_churn
5   where churnstatus = 'Churned' and PreferredOrderCat = 'Laptop & Accessory'
6   group by cityTier
7   order by count(*) desc

```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: [IA](#)

	cityTier	count(*)
▶	3	150
	1	60

➤ Identify the most preferred payment mode among active customers.

Limit to 1000 rows

```

1 • select * from customer_churn
2
3 ✖ select PreferredPaymentMode, count(*)
4   from customer_churn
5   where churnstatus = 'Active'
6   group by PreferredPaymentMode
7   order by count(*) desc

```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: [IA](#)

	PreferredPaymentMode	count(*)
▶	Debit Card	1956
	Credit Card	1522
	E wallet	474
	Cash on Delivery	386
	UPI	342

Result 53 x

➤ Calculate the total order amount hike from last year for customers who are single and prefer mobile phones for ordering.

Limit to 1000 rows

```
1 select sum(OrderAmountHikeFromLastYear) from customer_churn
2 where maritalStatus = 'Single' and PreferredOrderCat = 'Mobile Phone'
3
4
```

Result Grid

sum(OrderAmountHikeFromLastYear)
12177

➤ Find the average number of devices registered among customers who used UPI as their preferred payment mode.

Limit to 1000 rows

```
1 • select avg(NumberOfDeviceRegistered) from customer_churn where PreferredPaymentMode = 'UPI'
```

Result Grid

avg(NumberOfDeviceRegistered)
3.7174

➤ Determine the city tier with the highest number of customers.

Limit to 1000 rows

```

1 select * from customer_churn
2
3 select cityTier, count(*) as totalnumberofcustomer
4 from customer_churn
5 group by cityTier
6 order by count(*) desc
7

```

Result Grid

	cityTier	totalnumberofcustomer
▶	1	3666
	3	1720
	2	242

➤ Identify the gender that utilized the highest number of coupons.

```

3 SELECT
4 gender,
5 SUM(couponused) AS total_coupons
6 FROM customer_churn
7 GROUP BY gender
8 ORDER BY total_coupons DESC

```

Result Grid

	gender	total_coupons
▶	Male	5629
	Female	4034

➤ List the number of customers and the maximum hours spent on the app in each preferred order category.

```

3 SELECT
4 PreferredOrderCat,
5 COUNT(*) AS total_customers,
6 MAX(HoursSpentOnApp) AS max_hours_on_app
7 FROM customer_churn
8 GROUP BY PreferredOrderCat
9 ORDER BY total_customers DESC;

```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: |

	PreferredOrderCat	total_customers	max_hours_on_app
▶	Mobile Phone	2078	5
	Laptop & Accessory	2050	5
	Fashion	826	5
	Grocery	410	4
	Others	264	4

- Calculate the total order count for customers who prefer using credit cards and have the maximum satisfaction score.

```

1 select * from customer_churn order by HoursSpentOnApp desc
2
3 select customerID, sum(orderCount) as totalordercount
4 from customer_churn
5 where PreferredPaymentMode = 'Credit Card' and SatisfactionScore = (select max(SatisfactionScore) from customer_churn)
6 group by customerID
7 order by totalordercount desc

```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: |

	customerID	totalordercount
▶	53490	14
	54960	14
	50675	13
	52145	13
	54271	13

- What is the average satisfaction score of customers who have complained?

SQL Query:

```

3 select customerID, avg(SatisfactionScore)
4 from customer_churn
5 where complaintReceived = 'Yes'
6 group by customerID

```

Result Grid:

customerID	avg(SatisfactionScore)
50001	2.0000
50002	3.0000
50003	3.0000
50006	5.0000
50008	2.0000

Result 32

➤ List the preferred order category among customers who used more than 5 coupons.

SQL Query:

```

3 select CustomerID, PreferredOrderCat from customer_churn
4 where CouponUsed > 5

```

Result Grid:

CustomerID	PreferredOrderCat
50011	Others
50021	Laptop & Accessory
50022	Fashion
50086	Fashion
50094	Fashion

customer_churn 35

➤ List the top 3 preferred order categories with the highest average cashback amount.

➤ Find the preferred payment modes of customers whose average tenure is 10 months and have placed more than 500 orders.

➤ Categorize customers based on their distance from the warehouse to home such as 'Very Close Distance' for distances <=5km, 'Close Distance' for <=10km, 'Moderate Distance' for <=15km, and 'Far Distance' for >15km. Then, display the

churn status breakdown for each distance category.

➤ List the customer's order details who are married, live in City Tier-1, and their order counts are more than the average number of orders placed by all customers.

➤ a) Create a 'customer_returns' table in the 'ecomm' database and insert the following data:

ReturnID	CustomerID	ReturnDate	RefundAmount
1001	50022	2023-01-01	2130
1002	50316	2023-01-23	2000
1003	51099	2023-02-14	2290
1004	52321	2023-03-08	2510
1005	52928	2023-03-20	3000
1006	53749	2023-04-17	1740
1007	54206	2023-04-21	3250
1008	54838	2023-04-30	1990

b) Display the return details along with the customer details of those who have churned and have made complaints.